**IT 105 – Information Management I**

**Phase 4: Cloud Backup and Integration of Emerging Database Technologies**

**Reflection on Emerging Database Technologies**

At the start of developing our Inventory Management System, our main goal was simply to make sure the database worked correctly storing products, suppliers, and sales without errors. After exploring newer database technologies, we realized that functionality alone is not enough if the system is expected to handle growth and more complex tasks. Two emerging trends that stood out are distributed databases and ML-integrated databases, both of which could improve our project in different ways.

Distributed databases divide data across multiple servers instead of keeping everything in one place. This setup improves system reliability and performance, especially when handling large amounts of data or multiple users at the same time. For our inventory system, this could be useful if the business expands into multiple locations, since each branch could access the database efficiently without overloading a single server. However, this approach also introduces challenges. Managing multiple servers requires careful configuration, and maintaining consistency between them can be difficult. For a relatively small system like ours, this might add unnecessary complexity.

ML-integrated databases focus on making databases more intelligent by allowing them to perform basic machine learning tasks. In our case, this could help analyze sales data to predict which products will run low in stock or which items are in high demand. This would allow better planning and reduce the chances of overstocking or shortages. Despite these benefits, there are limitations to consider. Machine learning requires clean and well-organized data, and our current dataset may not be sufficient for accurate predictions. It also adds another layer of complexity that may not be practical for a beginner-level project.

Overall, these technologies show how databases are evolving beyond simple data storage. While our current system works well, integrating these innovations in the future could make it more scalable and intelligent, as long as the added complexity is carefully managed.

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